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Student Number

2023

YEAR 12
ASSESSMENT TASK 3

Mathematics Extension 2

Staff Involved:

DZP* BHC
Mr Peattie Mr

33 copies

8:30AM
WEDNESDAY 31 MAY

Carruthers

TOPICS COVERED

- Vectors / 25
- Integration / 25

General

Instructions:

- Working time – 55 minutes
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided separately
- Write all answers in the spaces provided
- Show relevant mathematical reasoning and/or calculations
- Marks may not be awarded for careless or poorly arranged working
- Diagrams are not to scale

Total marks:

50

Question 1

Marks

- (a) Find the vector parallel to $\vec{v} = 2\vec{i} - 3\vec{j} + 5\vec{k}$ which has a magnitude of 6.

2

- (b) (i) Find, in exact form, the angle between the position vectors:

2

$$\vec{OA} = 2\vec{i} - 3\vec{j} + 5\vec{k} \text{ and}$$

$$\vec{OB} = -\vec{i} + \vec{j} + 2\vec{k}.$$

- (ii) Hence find the exact area of triangle OAB.

1

(c) Consider the points $A (5, 6, 7)$ and $B (2, -1, 0)$.

(i) Find a vector equation of the line passing through A and B 2

(ii) Prove that the point $C (11, 20, 21)$ lies on the line passing through A and B. 1

(iii) Using (i), find a vector equation for the **interval** from A to B. 1

Question 2

(a) The points $A (9, 2, 5)$, $B (8, 3, W)$ and $C (10, 10, 9)$ form a right angled triangle, right angled at B.
Find the possible values of W . 3

(b) (i) What is the angle between the two vectors

$$\vec{v} = -\vec{i} + \sqrt{2}\vec{j} + \vec{k} \text{ and } \vec{u} = \vec{i} - \sqrt{2}\vec{j} - \vec{k} ?$$

1

(ii) What is the geometrical significance of the result in (i)? 1

- (c) Given $\vec{u} = 3\vec{i} + 3\vec{j} + 3\vec{k}$
 $\vec{v} = 3\vec{i} + 2\vec{j} - \vec{k}$
 $\vec{w} = -\vec{i} - \vec{j} - 2\vec{k}$
- (i) Show that $\vec{u} \circ \vec{v} = -\vec{u} \circ \vec{w}$

1

- (ii) Rearrange $\vec{u} \circ \vec{v} = -\vec{u} \circ \vec{w}$ so that the expression equals zero. Therefore which two vectors are perpendicular?

2

- (d) A parallelogram has all sides equal in length.
 Show, using vectors, that its diagonals are perpendicular.

3

Question 3

The equations $\frac{x}{2} = y - 2 = z - 2$ determine a straight line.

- (i) Write the equation of the line in vector form.

1

- (ii) This line intersects the sphere $x^2 + y^2 + z^2 + 2x + 2z = 4$ in two points.

Find the coordinates of these points.

4

Question 4

(a) Find $\int \frac{4x+3}{x^2+4x+5} dx$

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(c) Find $\int e^{2x} \sin x dx$

3

(b) Use the substitution $x = 2\cos \theta$ to find $\int \frac{dx}{\sqrt{(4-x^2)^3}}$

4

(d) Evaluate $\int_0^1 \frac{\sqrt{x}}{1+x} dx$ using an appropriate trigonometric substitution.

3

Question 5

Find $\int \frac{x-2}{(x^2+1)(x+1)} dx$

3**Question 6**

Let $A_n = \int_0^{\frac{\pi}{2}} \cos^{2n} x \, dx$, where n is an integer ≥ 0 .

Show that $nA_n = \frac{2n-1}{2} A_{n-1}$ for $n \geq 1$.

4

Question 7

Let $I_n = \int_0^{\frac{\pi}{4}} \sec^n x \, dx$, where n is an integer ≥ 0 .

(i) Show that $I_n = \frac{1}{n-1} [(\sqrt{2})^{n-2} + (n-2)I_{n-2}]$ for $n \geq 2$.

4

(ii) Hence evaluate I_4 .

1

END OF PAPER

Question 1

Marks

(a) Find the vector parallel to $\underline{v} = 2\hat{i} - 3\hat{j} + 5\hat{k}$ which has a magnitude of 6.

2

$$|\underline{v}| = \sqrt{4 + 9 + 25} = \sqrt{38}$$

$$\underline{\hat{v}} = \frac{1}{\sqrt{38}} \begin{pmatrix} 2 \\ -3 \\ 5 \end{pmatrix}$$

The required vector is $\frac{6}{\sqrt{38}} \begin{pmatrix} 2 \\ -3 \\ 5 \end{pmatrix}$

(b) (i) Find, in exact form, the angle between the position vectors:

1 2

$$\underline{OA} = 2\hat{i} - 3\hat{j} + 5\hat{k} \text{ and}$$

$$\underline{OB} = -\hat{i} + \hat{j} + 2\hat{k}.$$

$$\cos \theta = \frac{-2 - 3 + 10}{\sqrt{38} \cdot \sqrt{6}} = \frac{5}{\sqrt{38} \cdot \sqrt{6}}$$

$$\theta = \cos^{-1} \left(\frac{5}{\sqrt{38} \cdot \sqrt{6}} \right)$$

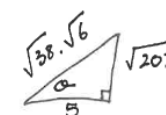
(ii) Hence find the exact area of triangle OAB.

2 1

Area of $\triangle OAB$

$$= \frac{1}{2} \times \sqrt{38} \times \sqrt{6} \times \frac{\sqrt{203}}{\sqrt{38} \cdot \sqrt{6}}$$

$$= \frac{1}{2} \sqrt{203} \text{ units}^2$$



(c) Consider the points A (5, 6, 7) and B (2, -1, 0).

(i) Find a vector equation of the line passing through A and B

$$\vec{AB} = \begin{pmatrix} -3 \\ -7 \\ -7 \end{pmatrix} \checkmark$$

$$\underline{r} = \begin{pmatrix} 5 \\ 6 \\ 7 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ -7 \\ -7 \end{pmatrix} \checkmark \text{ OR } \underline{r} = \begin{pmatrix} 5 \\ 6 \\ 7 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 7 \\ 7 \end{pmatrix}$$

$$\text{OR } \underline{r} = \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ -7 \\ -7 \end{pmatrix} \text{ OR } \underline{r} = \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 7 \\ 7 \end{pmatrix}$$

(ii) Prove that the point C (11, 20, 21) lies on the line passing through A and B.

$$\text{Does } \begin{pmatrix} 11 \\ 20 \\ 21 \end{pmatrix} = \begin{pmatrix} 5 \\ 6 \\ 7 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ -7 \\ -7 \end{pmatrix}?$$

Yes, when $\lambda = -2$ (or equivalent).

Hence the point C lies on the line.

(iii) Using (i), find a vector equation for the interval from A to B.

The interval from A to B is

$$\underline{r} = \begin{pmatrix} 5 \\ 6 \\ 7 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ -7 \\ -7 \end{pmatrix} \text{ for } 0 \leq \lambda \leq 1 \checkmark$$

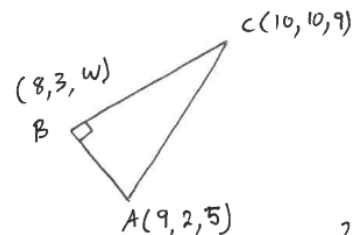
$$\text{OR } \underline{r} = \begin{pmatrix} 5 \\ 6 \\ 7 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 7 \\ 7 \end{pmatrix} \text{ for } -1 \leq \lambda \leq 0$$

$$\text{OR } \underline{r} = \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ -7 \\ -7 \end{pmatrix} \text{ for } -1 \leq \lambda \leq 0$$

$$\text{OR } \underline{r} = \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 7 \\ 7 \end{pmatrix} \text{ for } 0 \leq \lambda \leq 1$$

Question 2

(a) The points A (9, 2, 5), B (8, 3, W) and C (10, 10, 9) form a right angled triangle, right angled at B.
Find the possible values of W.



Now $\vec{BA} \cdot \vec{BC} = 0$ so

$$\begin{pmatrix} 1 \\ -1 \\ 5-W \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 7 \\ 9-W \end{pmatrix} = 0 \checkmark$$

$$2 - 7 + (5-W)(9-W) = 0$$

$$-5 + 45 - 5W - 9W + W^2 = 0 \checkmark$$

$$W^2 - 14W + 40 = 0$$

$$(W-10)(W-4) = 0 \checkmark$$

$$W = 4 \text{ or } 10.$$

(b) What is the angle between the two vectors

$$\underline{v} = -\underline{i} + \sqrt{2}\underline{j} + \underline{k} \text{ and}$$

$$\underline{u} = \underline{i} - \sqrt{2}\underline{j} - \underline{k}?$$

And what is the geometrical significance of this result?

$$\cos \theta = \frac{\begin{pmatrix} -1 \\ \sqrt{2} \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -\sqrt{2} \\ -1 \end{pmatrix}}{2 \times 2}$$

$$= \frac{-1 - 2 - 1}{4}$$

$$= -1$$

$$\therefore \theta = 180^\circ \checkmark$$

This means that the vectors point in opposite directions.

(c) Given $\underline{u} = 3\hat{i} + 3\hat{j} + 3\hat{k}$

$$\underline{v} = 3\hat{i} + 2\hat{j} - \hat{k}$$

$$\underline{w} = -\hat{i} - \hat{j} - 2\hat{k}$$

(i) Show that $\underline{u} \circ \underline{v} = -\underline{u} \circ \underline{w}$

$$\underline{u} \circ \underline{v} = 9 + 6 - 3 = 12$$

$$-\underline{u} \circ \underline{w} = 3 + 3 + 6 = 12$$

Hence $\underline{u} \circ \underline{v} = -\underline{u} \circ \underline{w}$

(ii) Rearrange $\underline{u} \circ \underline{v} = -\underline{u} \circ \underline{w}$ so that the expression equals zero. Therefore which two vectors are perpendicular?

$$\underline{u} \circ \underline{v} = -\underline{u} \circ \underline{w}$$

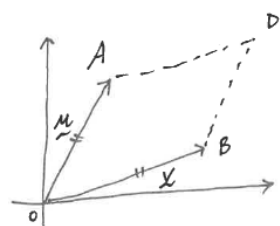
$$\underline{u} \circ \underline{v} + \underline{u} \circ \underline{w} = 0$$

$$\underline{u} \circ (\underline{v} + \underline{w}) = 0$$

$\underline{u}$ is perpendicular to $(\underline{v} + \underline{w})$

(d) A parallelogram has all sides equal in length.

Show, using vectors, that its diagonals are perpendicular.



$$\text{Let } \vec{OA} = \underline{u} \text{ and } \vec{OB} = \underline{v}.$$

$$\text{Then } \vec{OC} = \underline{u} + \underline{v} \text{ and}$$

$$\vec{BA} = \underline{u} - \underline{v}.$$

$$\text{Consider } \vec{OC} \circ \vec{BA}$$

$$= (\underline{u} + \underline{v}) \circ (\underline{u} - \underline{v})$$

$$= \underline{u} \circ \underline{u} + \underline{u} \circ \underline{v} - \underline{u} \circ \underline{v} - \underline{v} \circ \underline{v}$$

$$= \underline{u} \circ \underline{u} - \underline{v} \circ \underline{v}$$

$$= |\underline{u}|^2 - |\underline{v}|^2$$

$$= 0 \text{ since } OA = OB \text{ (data)}$$

Hence the diagonals are perpendicular.

Question 3

The equations $\frac{x}{2} = y - 2 = z - 2$ determine a straight line.

(i) Write the equation of the line in vector form.

$$\frac{x}{2} = \lambda \quad y - 2 = \lambda \quad z - 2 = \lambda$$

$$x = 0 + 2\lambda \quad y = 2 + \lambda \quad z = 2 + \lambda$$

$$\underline{r} = \begin{pmatrix} 0 \\ 2 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$$

(ii) This line intersects the sphere $x^2 + y^2 + z^2 + 2x + 2z = 4$ in two points.

Find the coordinates of these points.

The Cartesian equation of the sphere is $(x+1)^2 + y^2 + (z+1)^2 = 6$, with radius $\sqrt{6}$ units and centre at $(-1, 0, -1)$. Hence the vector equation of the sphere is

$$\left| \underline{r} - \begin{pmatrix} -1 \\ 0 \\ -1 \end{pmatrix} \right| = \sqrt{6}.$$

We need to solve

$$\left| \begin{pmatrix} 0 \\ 2 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} - \begin{pmatrix} -1 \\ 0 \\ -1 \end{pmatrix} \right| = \sqrt{6} \text{ i.e. } \left| \begin{pmatrix} 1+2\lambda \\ 2+\lambda \\ 3+\lambda \end{pmatrix} \right| = \sqrt{6}$$

$$\sqrt{1+4\lambda+4\lambda^2+4+4\lambda+\lambda^2+9+6\lambda+\lambda^2} = \sqrt{6}$$

$$6\lambda^2 + 14\lambda + 14 = 6$$

$$3\lambda^2 + 7\lambda + 4 = 0$$

$$(3\lambda + 4)(\lambda + 1) = 0$$

$$\lambda = -1 \text{ or } -\frac{4}{3}$$

Hence the points of intersection are $\begin{pmatrix} 0 \\ 2 \\ 2 \end{pmatrix} - 1 \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 2 \\ 2 \end{pmatrix} - \frac{4}{3} \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -\frac{8}{3} \\ \frac{2}{3} \\ \frac{2}{3} \end{pmatrix}$

Question 4

(a) Find $\int \frac{4x+3}{x^2+4x+5} dx$

$$\begin{aligned}
 &= \int \frac{2(2x+4)-5}{x^2+4x+5} dx \\
 &= 2 \int \frac{2x+4}{x^2+4x+5} dx - \int \frac{5}{1+(x+2)^2} dx \\
 &= 2 \ln|x^2+4x+5| - 5 \tan^{-1}(x+2) + C
 \end{aligned}$$

(b) Use the substitution $x = 2 \cos \theta$ to find

$$\int \frac{dx}{\sqrt{(4-x^2)^3}}$$

let $x = 2 \cos \theta$

$$\frac{dx}{d\theta} = -2 \sin \theta$$

$$I = \int \frac{-2 \sin \theta d\theta}{\sqrt{(4-4 \cos^2 \theta)^3}}$$

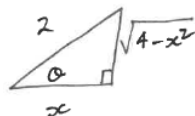
$$= -\frac{2}{8} \int \frac{\sin \theta d\theta}{\sin^6 \theta}$$

$$= -\frac{1}{4} \int \frac{d\theta}{\sin^5 \theta}$$

$$= -\frac{1}{4} \int \csc^5 \theta d\theta$$

$$= -\frac{1}{4} [-\cot \theta]$$

$$\cos \theta = \frac{x}{2}$$



$$= \frac{1}{4} \cot \theta$$

$$= \frac{1}{4} \times \frac{x}{\sqrt{4-x^2}}$$

$$= \frac{x}{4\sqrt{4-x^2}} + C$$

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(c) Find $\int e^{2x} \sin x dx$

3

$$\begin{aligned}
 &= \frac{1}{2} e^{2x} \sin x - \int \frac{1}{2} e^{2x} \cos x dx \\
 &= \frac{1}{2} e^{2x} \sin x - \left[\frac{1}{4} e^{2x} \cos x - \int \frac{1}{4} e^{2x} (-\sin x) dx \right] \\
 &= \frac{1}{2} e^{2x} \sin x - \frac{1}{4} e^{2x} \cos x - \frac{1}{4} \int e^{2x} \sin x dx \\
 \therefore \frac{5}{4} \int e^{2x} \sin x dx &= \frac{1}{2} e^{2x} \sin x - \frac{1}{4} e^{2x} \cos x \\
 \therefore \int e^{2x} \sin x dx &= \frac{4}{5} \left(\frac{1}{2} e^{2x} \sin x - \frac{1}{4} e^{2x} \cos x \right) \\
 &= \frac{e^{2x}}{5} (2 \sin x - \cos x) + C
 \end{aligned}$$

(d) Evaluate $\int_0^1 \frac{\sqrt{x}}{1+x} dx$ using an appropriate trigonometric substitution.

3

let $x = \tan^2 \theta$ then $\frac{dx}{d\theta} = 2 \tan \theta \sec^2 \theta$
when $x=1$, $\theta = \frac{\pi}{4}$ and when $x=0$, $\theta=0$. Hence

$$I = \int_0^{\frac{\pi}{4}} \frac{\sqrt{\tan^2 \theta} \times 2 \tan \theta \sec^2 \theta d\theta}{1 + \tan^2 \theta}$$

$$= \int_0^{\frac{\pi}{4}} 2 \tan^2 \theta d\theta$$

$$= 2 \int_0^{\frac{\pi}{4}} (\sec^2 \theta - 1) d\theta$$

$$= 2 [\tan \theta - \theta]_0^{\frac{\pi}{4}}$$

$$= 2 \left[1 - \frac{\pi}{4} - 0 \right] = 2 - \frac{\pi}{2}$$

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Question 5

Find $\int \frac{x-2}{(x^2+1)(x+1)} dx$

Let $\frac{x-2}{(x^2+1)(x+1)} = \frac{A}{x+1} + \frac{Bx+C}{x^2+1}$

then $x-2 = A(x^2+1) + (Bx+C)(x+1)$

When $x = -1$: $-3 = 2A$ $A = -\frac{3}{2}$

When $x = 0$: $-2 = A + C$
 $-2 = -\frac{3}{2} + C$ $C = -\frac{1}{2}$

When $x = 1$: $-1 = 2A + (B+C) \times 2$
 $-1 = -3 + 2B - 1$
 $3 = 2B$ $B = \frac{3}{2}$

$I = \int \frac{-\frac{3}{2}}{x+1} + \frac{\frac{3}{2}x - \frac{1}{2}}{x^2+1} dx$

$= -\frac{3}{2} \ln|x+1| + \frac{3}{4} \int \frac{2x}{x^2+1} dx - \frac{1}{2} \int \frac{dx}{1+x^2}$

$= -\frac{3}{2} \ln|x+1| + \frac{3}{4} \ln|1+x^2| - \frac{1}{2} \tan^{-1}(x) + C$

3

Question 6

Let $A_n = \int_0^{\frac{\pi}{2}} \cos^{2n} x dx$, where n is a positive integer.

Show that $nA_n = \frac{2n-1}{2} A_{n-1}$, for $n \geq 1$.

4

$A_n = \int_0^{\frac{\pi}{2}} \cos x \cdot \cos^{2n-1} x dx$
 $= \left[\sin x \cdot \cos^{2n-1} x \right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \sin x \cdot (2n-1) \cos^{2n-2} x \cdot (-\sin x) dx$
 $= 0 + (2n-1) \int_0^{\frac{\pi}{2}} \sin^2 x \cos^{2n-2} x dx$
 $= (2n-1) \int_0^{\frac{\pi}{2}} (1 - \cos^2 x) \cos^{2n-2} x dx$
 $= (2n-1) \int_0^{\frac{\pi}{2}} (\cos^{2n-2} x - \cos^{2n} x) dx$
 $= (2n-1) \int_0^{\frac{\pi}{2}} \cos^{2(n-1)} x dx - (2n-1) \int_0^{\frac{\pi}{2}} \cos^{2n} x dx$
 $= (2n-1) A_{n-1} - (2n-1) A_n$

$A_n + (2n-1) A_n = (2n-1) A_{n-1}$

$2n A_n = (2n-1) A_{n-1}$

$n A_n = \frac{2n-1}{2} A_{n-1}$

Question 7

Let $I_n = \int_0^{\frac{\pi}{4}} \sec^n x \, dx$, where n is a positive integer.

(i) Show that $I_n = \frac{1}{n-1} [(\sqrt{2})^{n-2} + (n-2)I_{n-2}]$ for $n > 1$.

4

$$\begin{aligned}
 I_n &= \int_0^{\frac{\pi}{4}} \sec^2 x \cdot \sec^{n-2} x \, dx \\
 &= \left[\tan x \cdot \sec^{n-2} x \right]_0^{\frac{\pi}{4}} - \int_0^{\frac{\pi}{4}} \tan x \cdot (n-2) \sec^{n-3} x \cdot (\sec x + \tan x) \, dx \\
 &= (\sqrt{2})^{n-2} - 0 - \int_0^{\frac{\pi}{4}} (n-2) \tan^2 x \sec^{n-2} x \, dx \\
 &= (\sqrt{2})^{n-2} - (n-2) \int_0^{\frac{\pi}{4}} (\sec^2 x - 1) \cdot \sec^{n-2} x \, dx \\
 &= (\sqrt{2})^{n-2} - (n-2) \int_0^{\frac{\pi}{4}} (\sec^n x - \sec^{n-2} x) \, dx \\
 &= (\sqrt{2})^{n-2} - (n-2) I_n + (n-2) I_{n-2} \\
 I_n + (n-2) I_n &= (\sqrt{2})^{n-2} + (n-2) I_{n-2} \\
 (n-1) I_n &= (\sqrt{2})^{n-2} + (n-2) I_{n-2} \\
 \therefore I_n &= \frac{1}{n-1} [(\sqrt{2})^{n-2} + (n-2) I_{n-2}], \text{ as required.}
 \end{aligned}$$

(ii) Hence evaluate I_4 .

1

$$\begin{aligned}
 I_4 &= \frac{1}{3} [(\sqrt{2})^2 + 2 I_2] \\
 I_2 &= \frac{1}{1} [(\sqrt{2})^0 + 0 \times I_0] = 1 \\
 \therefore I_4 &= \frac{1}{3} (2 + 2 \times 1) = \frac{4}{3}
 \end{aligned}$$

END OF PAPER